

# BOOK IV — GRAVITY / SPACETIME RESPONSE

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## 1 · The gravitational-response picture

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GRUT's proposal for gravity is constitutive rather than geometric-first: the gravitational vacuum is treated as a responsive medium, and what the metric perturbation does next depends on what has acted on it, through a retarded kernel carrying interaction history. The entire gravitational sector of the program is organized around one object — the retarded transverse-traceless (TT) response kernel  $K_R$  of the vacuum itself — written inside a single Schwinger–Keldysh influence action for the metric perturbation  $h$  (doubled fields, Keldysh basis  $h_r, h_a$ ):

$$S_{IF}[h_r, h_a] = \int (d\omega/2\pi) d^3k [ h_a \cdot K_R(\omega, k^2) \cdot h_r + (i/2) h_a \cdot N(\omega, k^2) \cdot h_a ]$$

with  $K_R = c_2 P^{(2)} + c_0 P^{(0)}$  decomposed on the Ward-surviving projector pair, subject to upper-half-plane analyticity, per-channel passivity, and the KMS/FDT lock tying the noise kernel  $N$  to  $\text{Im } \chi$  in equilibrium (GRUT\_MODEL\_FRAMEWORK.md §3).

Two honesty facts frame everything below, before any gravitational result is stated. First, the influence-functional form itself is generic:

**STATUS: RECOVERED (generic; u1: the form confers no GRUT-specific content)** — any local causal open system yields the Feynman–Vernon form; no GRUT credit accrues for the form (source: GRUT\_MODEL\_FRAMEWORK.md §3; canonical table item 1).

Second, the restriction of the vacuum's response to the pure spin-2 (TT) channel is a choice, not a theorem:

**STATUS: ASSUMPTION (STRUCTURAL-SELECTION — CHOSEN, unanimous five-angle interrogation; not forced)** — diffeomorphism invariance stops one condition short of forcing the TT projector; the {shear, bulk} interior remains the unexplored empirically viable sector (source: register node `p_tt_ansatz`; canonical table item 8).

And the equilibrium lock that disciplines the whole family:

**STATUS: ASSUMPTION (borrowed standard identity, enforced as a hard admission gate; rung2)** — KMS/FDT ties  $N$  to  $\text{Im } \chi$  by the coth factor; borrowed, not derived (source: GRUT\_MODEL\_FRAMEWORK.md §3; canonical table item 2).

The constitutive interpretation, stated at the level the record supports: GRUT does not claim to have derived a new gravitational dynamics. It claims that gravitational response, dissipation, and noise can be *written* as sector instances of one open-system architecture, and it then asks — through the campaigns recorded below — what that writing earns. The answer of the completed root-level audit is severe and is presented first-class in §6: the kernel that this architecture actually produces is produced by standard QFT/EFT acting on inputs GRUT declares rather than derives.

## 2 · The Tier-4 contract: the computed kernel

The backbone of the gravitational sector is the Tier-4 contract result (2026-09-01): the response formulation, run as an ordinary one-loop Schwinger–Keldysh computation on the declared inputs (two-

derivative Einstein–Hilbert TT-TT-TT cubic vertex; massless graviton bath; BD-analogue state; Option- $\beta$  dimensional continuation with no IR scale;  $k \rightarrow 0$ -first order of limits; the chosen TT projector), reproduces the standard one-loop retarded TT self-energy. At contract scope:

$$K_R(\omega) = \Sigma_R(\omega) \quad [(1/2 \kappa^2)\text{-weighted probe units}]$$

$$\Sigma_R(\omega > 0) = -(3/1280\pi^2) \omega^4 L + H^2(-(13/480\pi^2) \omega^2 L) + [\text{real local slot}],$$

$$L = \log(\mu^2/\omega^2) + i\pi, \quad H^1 \text{ sector} \equiv 0$$

with the real local polynomial ( $c_0, c_2, c_4, c_0', c_2'$ ) carried symbolically — the D5/scheme slot, never chosen — and the absorptive content exact:  $\text{Im } \Sigma_R^{\{H^0\}} = -(3/1280\pi)\omega^4$ ,  $\text{Im } \Sigma_R^{\{H^2\}} = -(13/480\pi)H^2\omega^2$ .

**STATUS: DERIVED (register tier SHOWN; declared scope:  $\omega \gg H$ ,  $k \rightarrow 0$ -first, reference-slice conditionals preserved verbatim)** — the contract-level retarded TT kernel exists and is well-defined in its declared domain; banked at ledger delta 0 as a scoped computed record, explicitly *not* a strength claim, with no derivation credit accruing to any GRUT rung (source: register node `kr_contract_retarded_tier4`; PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_RETARDED\_VERDICT.md).

The structural theorem inside it:

**STATUS: DERIVED (flat contract scope,  $\omega \gg H$ )** — Tier-4 TT kernel,  $H^1 = 0$  identically (source: canonical table item 5; PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_RETARDED\_VERDICT.md).

and its generalization,

**STATUS: DERIVED (the one genuinely new structural identity; carries no confirmatory weight for GRUT)** — the  $H^1 = 0$  four-channel cancellation and the even-degree ladder class (Theorem II) follow from exactly two standard identities; the true premise is evenness of momentum degree (source: canonical table item 24; memory-consolidated  $H^1$  closure package — the full derivation chain is Book II/X material).

The benchmark consequence run against the register's own expectations (2026-09-01) produced the sector's most important self-correction:

**STATUS: DERIVED (flat scope; rejects the framework's own registered  $s = 3$ )** — spectral law  $s = 5$ :  $J_{\text{eff}}(\omega) = (3/1280\pi)\omega^5 + (13/480\pi)H^2\omega^3$ ; the flat vacuum is a pure  $\omega^4$  power law in  $\text{Im } \chi$ , firmly convergent with  $\text{Re } \chi(0) = 3/(2560\pi^2)$  exactly; the registered  $s = 3$  power re-enters only as the curvature-induced  $O(H^2)$  component with an  $H^2$ -proportional coefficient the registered family excludes (source: canonical table item 6; PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_BENCHMARK\_VERDICT.md).

The benchmark's second axis (relaxational vs resonant) returned INDETERMINATE with the missing component precisely named — the D5 local/renormalization conditions; the registered test is scheme-hostage (the reference-slice sign crossing sits exactly at  $\omega = \mu$  and moves with  $\mu$ ). The scheme-robust substatement: for every real local choice there is no resonance-class feature in-domain —  $\text{Im } \chi$  monotone positive, no pole, no peak.

**STATUS: UNRESOLVED (axis 2 INDETERMINATE; missing component = D5 local conditions; row 4 of the registered cell as registered)** — no scheme-invariant relaxational/resonant adjudication exists on the current declarations (source: PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_BENCHMARK\_VERDICT.md ).

One further exact-de Sitter derived object belongs to this sector's inventory:

**STATUS: DERIVED (exact dS; gapped only at conformal coupling)** — the exact dS constant retarded tail  $H^2/4\pi$ , with the conformal gap at  $m_{\text{eff}}^2 = 2H^2$  (source: canonical table item 7; GRUT\_PROGRAM\_FREEZE.md §3).

### Validity boundary — derived, not declared

The kernel's validity domain is  $\omega \gg H$ , and its termination is itself a *result* of the calculation, not a stipulation. The expansion-control parameter is  $\varepsilon_H = (104/9)H^2/\omega^2$ , and the coefficient is exact arithmetic on the two computed absorptive coefficients:  $(13/480)/(3/1280) = 104/9$ . So  $\varepsilon_H \geq 1$  literally states "the second term of the H-expansion equals or exceeds the first," and the evaluator's refusal at  $\omega = \sqrt{(104/9)} \cdot H = 3.3993H$  is a consequence of the computed kernel — it cannot be loosened by declaration.

**STATUS: DERIVED (the refusal boundary  $\omega = 3.3993H$  is a result of the calculation: the 104/9 is the ratio of the two computed absorptive coefficients)** — ROOT-1 obstruction O1 (source: PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §3; PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_RETARDED\_VERDICT.md ).

The instrument enforces the boundary in hardware:  $\varepsilon_H \leq 0.1$  controlled, 0.1–1 returned only with an explicit BOUNDARY flag,  $\varepsilon_H \geq 1$  refused (DomainRejected) — and the refusal is itself a gated control.

### Ward status

The frozen T3 record's nonzero gauge-image contraction is carried through Tier 4 unchanged: the graviton-loop analogue of the Class-B structure persists, and  $K_R$  is TT-scoped by the frozen charter, so the residual is *excluded by construction* — not resolved, not repaired, and  $K_R$  was not altered to change it.

**STATUS: UNRESOLVED (Ward Class-B residual excluded by TT scope, not repaired)** — (source: PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_RETARDED\_VERDICT.md , T4-9).

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## 3 · Analytic structure: a cut, not the asserted pole

GRUT's historically central gravitational bet was a finite-memory, single-pole vacuum kernel. What the Tier-4 computation actually found at  $\omega = 0$  is a branch point with a real-axis cut — the gapless two-graviton continuum — present in both Dyson forms (first-order and resummed), one-loop in origin, unconditional within the contract.

**STATUS: DERIVED (contract scope; unconditional within the declared inputs)** — branch point at  $\omega = 0$  with a real-axis cut, the gapless two-graviton continuum; the cut is mandatory but input-forced (the ordinary massless two-particle threshold, from the declared massless bath) (source: PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_RETARDED\_VERDICT.md T4-8; PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §4).

No pole claim is made anywhere in the contract record. No residue is computed anywhere in the repository (grep-verified, zero hits); no width; no pole shown gauge-invariant. The absence-of-additional-zeros statement for the resummed denominator is triply conditional and frozen as such (reference slice  $c = 0$ ,  $\kappa = 0.1$  units,  $\mu = 1$ ); pole-from-cut candidates require  $(\kappa\omega)^2|\log| \sim 1$ , outside the EFT domain. Two naive inferences are refused in both directions by the record's own numbers: "cut  $\Rightarrow$  no pole" is false (matter-scope first-sheet poles coexist with a cut), and "TT projection precludes a pole" is false (the matter pole hunt ran in the TT sector); the matter-scope fence is respected and nothing is imported across it.

The status of the asserted kernel itself, verbatim from the canonical table:

**STATUS: UNRESOLVED (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope)** — single-pole/finite-memory kernel (rung3) (source: canonical table item 4; register node `rung3_single_pole`).

The decisive scope fact, which must travel with any quotation of "no pole": GRUT's claimed relaxation pole sits at  $\omega \sim H_0$  — *inside the region the evaluator refuses*. "No pole" is a statement about  $\omega \gg H$ ; the pole was never looked for where it was claimed (PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §4).

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## 4 · The infrared: $\omega \lesssim 3.4H$ is UNASKABLE, on four obstructions

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The region below the refusal boundary is not merely unresolved at current declarations — ROOT-1 §3 established that the question itself cannot currently be posed there. Four independent obstructions, each separately sufficient, presented faithfully:

- **O1 — the boundary is DERIVED, not declared.**  $\varepsilon_H = (104/9)H^2/\omega^2$  with 104/9 exactly the ratio of the two computed absorptive coefficients (§2 above). The refusal at  $\omega = 3.3993H$  is a result of the calculation and cannot be loosened by declaration.
- **O2 — there is no frequency variable there.** The background time-translation flow is a priced assumption (register node `background_time_translation_flow`, tier assumed, +1); without the flow "there is no single- $\omega$  kernel and no  $\omega \rightarrow 0$  transport coefficient to conjecture about." Its only named discharge route (the static patch) is exhibited false for the TT graviton, with shapes differing up to 134% across epochs.  $\rho(\omega)$ ,  $J(\omega)$ ,  $\text{Im } \chi(\omega)$  are not defined objects at  $\omega \lesssim H$ ; any low-frequency spectral claim — convergent or floored — presupposes exactly what is missing.
- **O3 — an unregulated IR log at  $O(H^2)$ ,** coefficient  $-8\omega^2/15$  per  $H^2$  at  $d = 3$ ; nine candidate regulators were swept, zero licensed, none adopted.

- **O4 — the declared scheme is provably blind to the effect it would adjudicate.** The candidate IR floor is thermal, but de Sitter thermality is invisible to the H-grading:  $\exp(-2\pi\omega/H)$  vanishes to all orders in  $H$ , and the blindness was *derived* ( $\coth(\pi\omega/H) - 1$  has vanishing limit and vanishing first and second  $H$ -derivatives at fixed  $\omega > 0$ ), not asserted.

**STATUS: UNRESOLVED (UNASKABLE at current declarations — four independent obstructions, each separately sufficient; not false)** — the  $\omega \lesssim 3.4H$  regime (source: PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §3; GRUT\_PROGRAM\_FREEZE.md §3 UNRESOLVED list).

Two standing guards travel with this verdict. First, the record refuses in advance any future "we computed it and there is no floor" produced from a graded calculation — the scheme can return neither a floor nor its absence, so such a gate would be definitional in the adverse direction. Second, the "new scale required" framing is refused: GRUT already has the number (rung2 records that Hadamard/KMS forces  $T_{dS} = H/2\pi$  uniquely, given the background). What is missing is a *licence* to use it, a *method* non-perturbative in  $H$ , and a *proved stationary reduction* — without which the low-frequency question has no referent.

**STATUS: DERIVED (within declarations: forced by Hadamard/KMS on the declared background)** —  $T_{dS} = H/2\pi$  (source: canonical table item 3; register node `rung2_kms_gate`).

## 5 · The object-identity fork: kernel vs dressed response — OWNER FORK

Every classification-bearing verdict in the gravitational sector was read on  $\chi = -K_R$  — the *undressed* 1PI kernel. But the pre-registration's pipeline and the Class-C manifest both name the *dressed* object  $G_R = 1/(G_0^{-1} - \Sigma)$ , and the two readings land on opposite sides of the pre-registered convergence boundary: the dressed reading gives  $\lim \text{Im } G_R = -3\kappa^4/(320\pi)$  on the reference slice — Ohmic and log-divergent, the side that trips rung1's own falsifier. The benchmark verdict carries the sensitivity disclosure explicitly: the kernel-level reading is load-bearing, and if the owner rules the registered object is the dressed response, the axis verdicts must be re-run on it. The divergence between the readings is contingent on  $c_0 = 0$ , which the D5 execution has since made exact — so the fork may already be decided in the adverse direction by a result the record has not caught up with. No document carries the fork forward in either direction (zero "dressed" mentions across all four 2026-09-03 audits).

**STATUS: UNRESOLVED (owner fork, still owed; the dressed reading lands on the opposite side of the pre-registered convergence boundary)** — kernel-vs-dressed object identity; this book does not resolve it and neither reading is adopted here (source: PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §11; PHYSICS\_LEDGER/WALL\_KR\_CONTRACT\_BENCHMARK\_VERDICT.md "Limitations"; GRUT\_PROGRAM\_FREEZE.md §3).

## 6 · Where the kernel comes from: the KERNEL-STANDARD verdict

The root-level question — *does anything internal to GRUT, as opposed to a declared input or standard EFT/QFT machinery, select the GRUT response kernel?* — was adjudicated by the ROOT-1 campaign (2026-09-04; primary plus independent adversarial leg, 42/42 battery, agreeing verdicts). The answer is NO: *why this kernel* — *because standard QFT/EFT produces it*, acting on inputs GRUT declares rather than derives. Not one link of the chain (vertex, bath, state, scheme, order of limits, projector) is a GRUT principle. All nine variation axes produced a countermodel; nine rescue principles classified as five category-2 (standard QFT/EFT) and four category-5 (ad hoc), zero category-4 (genuinely new GRUT principle) — and the gate was two-sided, so a single category-4 would have flipped it.

**STATUS: CLOSED (gate outcome: KERNEL-STANDARD — primary and adversarial leg agree; nothing internal to GRUT selects the kernel)** — with the mandatory scope warning adopted from the adversarial leg: KERNEL-STANDARD means *the  $\omega \gg H$  stand-in is accounted for by standard physics*; the kernel GRUT actually asserts (finite memory, single pole,  $s = 3$ ) lives at  $\omega \lesssim H$  and was never computed there — on the claimed object nothing determines the kernel, not GRUT and not standard EFT (source: PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §§1–2, scope box, §13).

Two structural barriers, GRUT's own, explain why nothing internal ever could select it: the framework asserts a class, and a class has no scale; and the admissible set is an amplitude-homogeneous cone that orients channels and pins no ratio — "every route from this framework to a number runs outside it" (docs/WHERE\_IT\_STOPS.md, as quoted in ROOT-1 §6).

The strongest successful countermodel deserves its own line, because it names an undeclared input: a Wilsonian system/bath partition at a comoving scale  $q_s$ , holding every declared input fixed, multiplies the  $H^0$  absorptive part by  $\theta(\omega - 2q_s)$  — turning the unconditional gapless branch point into a gapped threshold. It is admissible because the contract's declaration sheet (D1–D5) never declares the system/bath mode partition at all.

**STATUS: UNRESOLVED (a load-bearing input — the system/bath mode partition — is undeclared in D1–D5; the gaplessness of the cut is contingent on it)** — (source: PHYSICS\_LEDGER/ROOT1\_KERNEL\_ORIGIN.md §8, §12).

## 7 • Known recoveries and limits

### 7.1 GR at zero-memory collapse

**STATUS: RECOVERED (largely by identity; KERNEL-STANDARD, scoped)** — GR at zero-memory collapse (source: canonical table item 19; GRUT\_PROGRAM\_FREEZE.md §3 RECOVERED).

The register's own record of the mechanism is blunt (node `rung5_gr_limit`, tier assumed, ledger +2):  $\tau_c \rightarrow 0$  collapses  $\chi$  to its conservative local form, and the Einstein equations are then recovered via the Clausius/Rindler-horizon route — which is Jacobson's argument, not GRUT's. The in-in formalism does no work in that step. The recovery imports two declared items (entropy  $\sim$  area, which sets  $G$ ; Unruh temperature, which sets  $\hbar$ ) and is priced +2 *because* it is a recovery: the specialist verdict on record is that the diffeomorphism Ward identity enforces only  $\nabla_{\mu} T^{\mu\nu} = 0$  and does not determine the action

— EH,  $R^2$ ,  $f(R)$ , Lovelock, and nonlocal curvature actions all satisfy the same Ward identities — so GR is *borrowed, not hosted*.  $\Lambda$  remains undetermined by the framework.

**STATUS: ASSUMPTION (register tier assumed; +2 imports: entropy~area and Unruh T; a recovery that must not be sold as a derivation)** — the zero-memory GR limit's mechanism (source: register node `rung5_gr_limit`, `tier_note` and `ledger_note`).

Recovery-by-identity is compatibility, not correspondence evidence (`GRUT_PROGRAM_FREEZE.md` §3). Nothing in this subsection counts toward PREDICTED.

## 7.2 Lorentz covariance of the response — priced, then discharged

The claim that the vacuum response kernel belongs to the Lorentz-covariant subspace was owner-ruled a priced assumption (+1) on 2026-08-24, with a measured basis: the FRW gauge-allowed bilinear response space is 11-dimensional, the Lorentz-covariant flat-compatible subspace is 3-dimensional, and the 8 removed structures were explicitly computed. It was then discharged by its own clause on 2026-08-30: the microscopic  $\Sigma_R^{\text{TT}}$  calculation demonstrated the response lies in the 3-dimensional subspace *without imposing it as an input* ( $Q1^{\text{TT}} = \text{INSIDE}$ : the nonlocal TT block equals  $a(\omega, k, H, m) \cdot P_2^{\text{TT}}$  exactly at  $H^0/H^1/H^2$ ; predicates hash-frozen before the TT numbers existed;  $Q5^{\text{TT}} = \text{INSIDE}$ : flat limit matches placement), and the +1 was retired.

**STATUS: DERIVED (discharged 2026-08-30: the priced response property demonstrated unimposed, +1 retired; scope  $\omega \gg H$ ;  $c_0 = 0$  remains separately unlicensed)** — Lorentz covariance of the vacuum response kernel (source: register node `response_lorentz_covariance`, `sub_status` and `ledger_note`; `PHYSICS_LEDGER/OWNER_ADJUDICATION_WALL_A_CLOSURE.md` as cited there).

This is the sector's cleanest example of the program's discipline working as designed: an assumption priced on entry, given a falsifier, and retired only when a computation met the node's own discharge condition.

## 7.3 Solar-system safety via cutoff suppression

The vacuum's dissipative response is Planck-cutoff suppressed, and the effect *grows* with frequency — so solar-system tests (orbital  $\mu\text{Hz}$ – $\text{mHz}$ ) are even more suppressed than the LIGO band, where the effect is already 22–62 orders below detectability (§8.2). There is no regime where the GW effect is large while solar-system physics is safe; both are suppressed by the same structure, GW less so.

**STATUS: DERIVED (scaling/consistency argument within the rung4 computation: the same structural Planck suppression that makes the GW effect invisible underwrites solar-system safety; no dedicated solar-system bound computation exists in the record)** — (source: `calc/RESULTS_gw.md` "Consistency"; `calc/gw_dissipation_bounds.py` header; register node `rung4_love_kk` `tier_note`).

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## 8 · Gravitational-wave physics

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## 8.1 The Love-number / Kramers–Kronig leg (rung4)

The elastic side of the vacuum's response —  $\text{Re } \chi$ , the storage modulus — is the vacuum analogue of a Love-number (tidal) response, Kramers–Kronig-linked to the dissipative  $\text{Im } \chi$ ; the structure recovers the worldline-EFT tidal-response organization for the vacuum.

**STATUS: RECOVERED (register tier SHOWN; borrowed KK/worldline-EFT tidal-response structure on the declared inputs)** —  $\text{Re}[\chi]$  = elastic/storage (Love-number) response, KK-linked to dissipative  $\text{Im}[\chi]$  (source: register node `rung4_love_kk`, statement and tier).

The static-transfer question — whether the per-channel *dissipative* sign floor ( $\omega \cdot \text{Im } \chi \geq 0$ ) transfers to the static/quasi-static modulus that observable couplings actually see — was posed into the register as its own node and answered in 2026-08-09 at the two unflattering outcomes jointly: the decomposition identity  $\chi(0) = \chi_\infty + (2/\pi) \int_0^\infty d\omega \text{Im } \chi / \omega$  shows passivity pushes the static modulus up *from* the instantaneous part  $\chi_\infty$  and never below it, so the whole transfer question is the sign of  $\chi_\infty$  — a reactive contact/UV datum that passivity, causality, and the KMS lock are all structurally blind to. A banked counterexample (a retarded-analytic, pointwise-passive, KMS-consistent kernel with  $\chi(0) = -0.6 < 0$ ) refutes unconditional transfer permanently;  $\chi_\infty \geq 0$  is the tightest sufficient premise, and whether GRUT's vacuum kernel satisfies it is bath/UV structure — rung3's domain, priced there.

**STATUS: UNRESOLVED (register tier derived-pending; answered at outcomes (ii)+(iii): unconditional sign transfer REFUTED permanently; the conditional floor gates on the sign of  $\chi_\infty$ , a contact/renormalization datum)** — the dissipative-to-static sign transfer (source: register node `kk_static_transfer`, `sub_status` and `boundary_condition`; `calc/kk_static_transfer.py`).

## 8.2 GW dissipation: real, and 22–62 orders too small

A finite dissipative  $\text{Im } \chi$  is something a lossless GR vacuum does not have: it would give a propagating GW frequency-dependent attenuation, dephasing, and  $v_g(\omega) \neq c$ . The rung4 computation (2026-06-25, `calc/gw_dissipation_bounds.py`) asked the honest question — not "extract a bound" but "is the effect within many orders of sensitivity?" — with the coupling set to 1, most generous for detectability. Outcome (B): real-but-unobservable.

- Predicted accumulated dephasing over 40 Mpc at 100 Hz:  $4.4 \times 10^{-23}$  rad ( $q = 1$  thermal reading) to  $1.5 \times 10^{-63}$  rad ( $q = 2$ ), against a  $\sim 0.1$  rad detectability threshold — 22 to 62 orders below, across the whole 10–1024 Hz band.
- GW170817 speed bound  $|c_{\text{gw}} - c|/c < 10^{-15}$ : satisfied with 26–66 orders to spare — not binding.
- The live window for  $|\chi|$  is  $[8 \times 10^{-20}, 2 \times 10^{-15}]$ ; GRUT sits 21–62 orders below it, and reaching it would require dropping the vacuum cutoff to MeV–meV scales — grossly excluded by particle-physics and equivalence-principle tests. You cannot tune into detectability without breaking everything else; the smallness is structural.

**STATUS: CLOSED (computed outcome B — real-but-unobservable; FAILS-DIFFERENTIATION; ledger 0)** — GW dissipation from  $\text{Im } \chi$ : real (absent in lossless GR) but  $\sim 22\text{--}62$  orders below LIGO-class sensitivity; Planck suppression structural, not tuned (source: register node `rung4_love_kk_tier_note`; `calc/RESULTS_gw.md`; `calc/gw_dissipation_bounds.py`; SIGNATURE\_AUDIT.md table row 1).

A scope fence added 2026-08-20 travels with the dephasing statements and is reproduced faithfully: the  $|\text{Re } \chi| \sim |\text{Im } \chi|$  step behind the "same order" amplitude-loss line holds only on the power-law branch and is false once a second (IR) pole is present — a Lorentzian pole does not switch off above its own frequency; for the two-scale kernel the register books, the UV/IR crossover lands at the geometric mean and moves  $\sim 10$  orders on an unpinned constant  $\omega_c$  (three in-corpus values spanning 39.6 orders). The  $\Delta\phi$  dephasing numbers are computed from  $\text{Re } \chi$  on the power-law branch and are unaffected — the filed "22–62 orders below" *stands as a dephasing statement* — but the amplitude channel of a second IR pole is not covered by anything computed in that file, and the IR pole's achromatic friction contribution is exactly the  $\Gamma_T$  slot of §8.5.

**STATUS: UNRESOLVED (the amplitude channel of a two-scale kernel is not covered by the rung4 computation; the crossover rides an unpinned  $\omega_c$  across  $\sim 10$  orders)** — (source: `calc/gw_dissipation_bounds.py` header, REGIME CHECK block, corrected 2026-08-20).

### 8.3 The graviton-mass entry — demoted to vacuous

The audit's graviton-mass line (a dS-horn  $m \sim H_0$  against the LVK dispersion bound) was demoted in the 2026-08-02 verification: the dispersion observable scales as  $m^2$ , so the induced phase of any  $m \sim H_0$ -class mass is  $10^{-21}$  of threshold — *a test that was never capable of running, not a test passed*. Against the best current bound (CMB dipole,  $5 \times 10^{-32}$  eV) the margin is  $\sim 1.5$  orders, not 10; and any  $m \sim H_0$ -class mass has Compton wavelength beyond the Hubble radius — of no observational significance in principle (Trivedi–Loeb). The  $10^{-21}$  figure is demotion arithmetic about a test's reach and is fenced against ever being re-derived into "GRUT's effect size" — no such calculation exists.

**STATUS: CLOSED (demoted to vacuous — arithmetically right, evidentially empty; the  $\leq 10^{-21}$  figure is a test-reach number, hard-fenced against banking as a GRUT effect)** — (source: SIGNATURE\_AUDIT.md item 3 and the NOT-banked fence).

### 8.4 The category distinction: $\alpha_M$ is not $\text{Im } \Sigma_R$

Standard-siren friction  $\alpha_M$  (running Planck mass) comes from a Hermitian action: removable by field redefinition, graviton-number-conserving, sign-indefinite, achromatic, noiseless. A genuine dissipative  $\text{Im } \Sigma_R$  is none of those. The two are slot-degenerate in the mean-field tensor equation only. Consequences carried on the record: a detected  $\Xi_0 \neq 1$  could never *confirm* GRUT, and would bear on GRUT only after a conservative-vs-dissipative decomposition the current parameterization does not perform.

**STATUS: DERIVED (audit-level category distinction within standard theory; no GRUT credit — the distinction identifies the dissipative class, not GRUT within it)** —  $\alpha_M \neq \text{Im } \Sigma_R$  (source: SIGNATURE\_AUDIT.md item 4; register node `rung3_single_pole_category_fence`).

The one measured statement in this sector is external: the Salcedo–Colas–Dufner–Pajer open EFT (JHEP 02(2026)241, arXiv:2507.03103) parameterizes a genuine dissipative tensor friction  $\Gamma_T$  with a mandatory stochastic source, explicitly distinct from  $\alpha_{M\cdot H}$  — the first mainstream parameterization of the object *class* — and via slot-degeneracy current friction measurements transfer as  $|\Gamma_T| \lesssim \text{few} \times H_0$  on the *shared* slot: a no-cancellation bound on the slot, not a decomposed measurement of GRUT's kernel.

**STATUS: ASSUMPTION (EMPIRICAL-INPUT, external, shared-slot; the retarded+noise form itself is u1-generic — no validation credit for the form) —** the SCDP  $\Gamma_T$  slot bound (source: SIGNATURE\_AUDIT.md item 5; arXiv:2507.03103 as the record cites it).

## 8.5 The $\Gamma_T$ closure — summary

$\Gamma_T$  was the currently identified candidate for a nontrivial cross-sector consequence — the one slot connecting the derived Tier-4 dissipation kernel, the KMS lock, and cosmological tensor propagation. The owner-ordered design gate (2026-09-06) found at design time, from register statuses alone, that the design does not survive as a prediction hunt; the subsequent closure computation (`calc/gw_tensor_friction.py`, same date) executed the SPEC and returned REFUSE on the observable route with the obstruction stack named (`p_tt CHOSEN`; frame-level CHOSEN;  $\tau_2$  inserted un-sourced; the  $\omega \approx 3.4H$  region UNASKABLE), and computed the only parameter-free entry — the Tier-4 derived chromatic friction  $\Gamma_T(\omega) = (3/1280\pi)(\omega^3/M_{P^2})[1 + (104/9)(H_0/\omega)^2]$ ,  $\mu$ -independent since  $\text{Im } L = \pi$ :

$\Gamma_T/H_0 = 6.19 \times 10^{-6.3}$  at 100 Hz — 62.7 orders below the shared-slot bound  $\text{few} \times H_0$ ,

final at these declarations because the horn has no free parameter. Every observable-sized route runs through inserted, staked, or choice-dependent inputs.

**STATUS: CLOSED (computed NO EFFECT; SPEC outcome REFUSE on the observable route; commits 2116251, 41e1af5) —**  $\Gamma_T$  parameter-free value ( $6.19e-63 \cdot H_0$  at 100 Hz) (source: canonical table item 16; GRUT\_PREDICTION\_GATE\_GAMMA\_T.md; `calc/RESULTS_gw_tensor_friction.md`). Cross-reference: Book IX carries the full gate story — the eleven-step protocol, the decision tree, and the standing consequence that no discriminator is identified on the current record.

## 8.6 The one soft spot: black-hole ringdown / QNM

Exactly one GW observable was not closed by a dedicated calculation: black-hole quasinormal modes / ringdown damping, where a lossy tidal response could in principle differ from GR's conservative one. The expectation on record is that any shift inherits `rung4`'s structural Planck suppression — but this is an inheritance argument, not a computation, and the signature audit's EMPTY verdict carries this one explicit caveat.

**STATUS: UNRESOLVED (invisible-by-inheritance, not a dedicated calculation — the signature audit's single explicit caveat) —** QNM/ringdown (source: SIGNATURE\_AUDIT.md "The one soft spot").

## 9 · What this sector does not contain — the absence map

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Stated explicitly, because the record is silent and silence is content:

- **Strong-field gravity.** No GRUT account of black-hole interiors, horizons as response structures, gravitational collapse, or binary dynamics/GW generation exists in the record. The only strong-field-adjacent item is the QNM inheritance caveat (§8.6). **STATUS: UNMAPPED** (no register node; no calc).
- **Nonlinear gravitational response.** The contract is one-loop, linear-response, TT-only, at the  $k_{\text{ext}} = 0$  evaluation point of the D1 limit;  $\Sigma$ 's  $O(k^2)$  is not computed (T3 scope, disclosed). **STATUS: UNMAPPED beyond the declared scope** (source: `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md` "Scope limitations").
- **The interacting graviton zero-mode (O2).** Undone; it decides the referent of the persistence fixed point and is reopening condition 1 of the program freeze. **STATUS: UNRESOLVED (posed, decidable, unrun)** (source: `GRUT_PROGRAM_FREEZE.md` §5).
- **The {shear, bulk} interior.** The scalar-sector response foreclosed by the chosen TT projector is empirically viable and windowed by data, not deleted by derivation — it is Book VI/IX territory but its existence conditions every "TT-only" statement in this book (source: `SIGNATURE_AUDIT.md` post-interrogation note; register node `zeta_interior_family`).
- **The noise-sector fork ( $\alpha = -2$ ) and D5's frozen renormalization conditions.** Both owner-held, both untouched by the Tier-4 record by construction; the D5 locals are the named missing component of the benchmark's axis 2 (source: `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md` HARD STOP; `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`).

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## 10 · The sector in one paragraph

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The gravitational sector is the program's most fully executed front and its most disciplined self-audit. What it established: the response formulation reproduces the standard one-loop retarded TT self-energy on the declared inputs (SHOWN, declared scope), with a genuinely new structural identity inside it ( $H^1 = 0$  and its even-degree ladder class, DERIVED — carrying no confirmatory weight for GRUT), a derived spectral law  $s = 5$  that rejects the framework's own registered  $s = 3$ , and a derived validity boundary at  $\omega = 3.3993H$  below which the record's own instruments refuse to answer. What it found against the founding bet: a cut, not the asserted pole — with the claimed pole never looked for where it was claimed, because at current declarations it *cannot* be looked for there. What it recovered: GR at zero-memory collapse, largely by identity, with its imports priced. What it predicts observationally: nothing — GW dissipation is real and 22–62 orders too small; the graviton-mass test was never capable of running; the one parameter-free number ever computed ( $\Gamma_T$ ) landed 62.7 orders below the only bound on its slot and closed its own gate. The kernel at the center of it all is, by the record's own two-sided adjudication, standard physics on declared inputs — and the question the sector hands to the rest of the corpus is not "why this kernel" (answered: standard QFT/EFT) but the one ROOT-1 names as deepest: *what is the bath, and by what partition is it separated from the system?*

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## Sources drawn from

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- `books/CORPUS_CHARTER.md`
- `GRUT_MODEL_FRAMEWORK.md`
- `GRUT_PROGRAM_FREEZE.md`
- `PHYSICS_LEDGER/ROOT1_KERNEL_ORIGIN.md`
- `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`
- `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md`
- `provenance/claims.json` — nodes `kr_contract_retarded_tier4`, `rung4_love_kk`, `rung5_gr_limit`, `rung3_single_pole`, `kk_static_transfer`, `response_lorentz_covariance`, `p_tt_ansatz` (via charter table and audits), `rung2_kms_gate` (via charter table)
- `calc/gw_dissipation_bounds.py` (header + REGIME CHECK block)
- `calc/RESULTS_gw.md`
- `calc/RESULTS_gw_tensor_friction.md`
- `GRUT_PREDICTION_GATE_GAMMA_T.md`
- `SIGNATURE_AUDIT.md` (table; items 3–5; NOT-banked fence; QNM soft spot)
- `docs/WHERE_IT_STOPS.md` (as quoted in ROOT-1 §6)
- External literature only as the record cites it: arXiv:2507.03103 (SCDP open EFT); arXiv:2103.08547, arXiv:2107.13905, arXiv:1307.1422, arXiv:2602.07908 (dS graviton self-energy line, via `rung3_single_pole`).

## Gaps in this book

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1. **The  $\omega \approx 3.4H$  regime has no content here because the record has none** — presented as UNASKABLE on four obstructions; any future account requires a licence, a method non-perturbative in  $H$ , and a proved stationary reduction (ROOT-1 §3).
2. **The kernel-vs-dressed fork is presented, not resolved** — every quantitative statement in §§2–3 and §8 is on the undressed reading and would need re-running if the owner rules for the dressed object.
3. **No strong-field content:** black holes, collapse, binary dynamics, GW generation — UNMAPPED in the record; only the QNM inheritance caveat exists.
4. **No dedicated QNM/ringdown calculation** — the audit's one explicit caveat stands.
5. **No dedicated solar-system bound computation** — safety is a scaling/consistency argument riding the same structural suppression as the GW smallness.
6. **The GW amplitude channel of a two-scale kernel is uncovered** — the 2026-08-20 fence; the crossover rides an unpinned  $\omega_c$  across  $\sim 10$  orders (three in-corpus values span 39.6 orders; the adjudication is owed). The  $\sim 10$ -order figure and Book IX's 19.8-order span are consistent, not competing; the crossover scales as  $\sqrt{\omega_c}$  (`calc/SPEC_gw_tensor_friction.md` §4), so the full three-value span halves to 19.8 orders of crossover while an adjacent-pair choice moves it  $\sim 10$ .
7. **D5's frozen renormalization conditions and the noise-sector ( $\alpha = -2$ ) fork** — both owner-held; axis 2 of the benchmark is INDETERMINATE until D5 lands.

8.  $\Sigma$ 's  $O(k^2)$  is not computed ( $k_{\text{ext}} = 0$  evaluation point; T3 scope), and the Ward Class-B residual is excluded by TT scope, not repaired.
9. The system/bath mode partition is undeclared in D1–D5 — a load-bearing input named by ROOT-1's strongest countermodel; the gaplessness of the cut is contingent on it.
10. O2 (interacting graviton zero-mode), the RESIDUE and SLOT tests — posed, decidable, unrun; held as the freeze's reopening keys, not as this book's content.
11. The  $\Gamma_T$  gate story is summarized, not reproduced — the full eleven-step protocol, decision tree, and conditional exhibits live in Book IX's scope.